# Fixed-Geometry In-Context Linear Sampling with Learned Acquisition Design

A two-dimensional proof of concept

August 4, 2026

**Abstract**

We test a fixed-feature in-context architecture on the actual linear sampling method (LSM), rather than on one-dimensional Gaussian-process regression. Each task is a two-dimensional active multistatic inverse-scattering problem with a complex far-field matrix. A prescribed angular softmax kernel defines the GP feature geometry. A tied complex preconditioned conjugate-gradient (PCG) loop then solves all Bayesian LSM sampling equations in context. Only a positive preconditioner built from the fixed attention geometry is trained; no temperature, image surrogate, indicator, noise estimator, or spatial regularization map is learned. Across three independent seeds and 576 held-out tasks per condition, the 20-loop model matches direct Tikhonov localization at 15% relative noise, including unseen kites and two-obstacle configurations, while attaining mean relative sampling-equation residuals between $1.0 \times 10^{-2}$ and $1.8 \times 10^{-2}$. A controlled recurrent-solver ablation shows that heavy-ball and Chebyshev iterations can localize at the same depth, but leave residuals above 0.54; Richardson requires about 80 loops even for localization. In a second proof of concept, six continuous source/receiver angles are trained end to end. The learned design reaches average precision 0.958/0.963/0.879 on ellipses/kites/two obstacles, versus 0.260/0.263/0.244 for six uniformly spaced angles. A point-spread audit attributes this gain to suppression of periodic grating lobes. These results support the mechanism and experiment-design idea, but remain a synthetic Born–Foldy study rather than a full sound-soft validation.

## 1 Question tested and precise scope

The experiment asks whether a prescribed nonlinear kernel geometry can be combined with a tied in-context loop to solve LSM systems, and whether the same differentiable pipeline can optimize an acquisition geometry. It does *not* learn a map from a response matrix to an obstacle image.

The physical regime is two-dimensional, active, deterministic and multistatic. There are 32 fixed plane-wave incident directions and 32 fixed receiver directions on a full circle. The randomness across tasks lies in the obstacle and measurement noise, not in the sources. Consequently this first proof of concept uses the far-field matrix $F_D$ and is distinct from the random-source cross-correlation matrix $C_D$ in Garnier–Haddar–Montanelli [1] and the mean-removed matrix $\widetilde{C}_D$ generated by a moving small random scatterer [2].

It is also distinct from Chenu–Haddar–Montanelli [3]. Their neural operator learns an indicator-dependent regularization function before a classical LSM solve. Here the indicator is never represented by a neural network: the recurrent cell itself executes the sampling solve, and the nonlinear feature kernel is fixed by construction.

## 2 Two-dimensional LSM task

Let $\widehat{x}_r, d_s \in \mathbb{S}^1$ be receiver and incident directions and let $D_h$ be the grid representation of an obstacle. The fast training-only forward model is the Born–Foldy sum

$$(F_D)_{rs} = \frac{1}{|D_h|} \sum_{y \in D_h} \exp\{ik(d_s - \widehat{x}_r) \cdot y\} + E_{rs}, \qquad F_D \in \mathbb{C}^{m \times m}. \tag{1}$$

For every probing point $z \in \Omega \subset \mathbb{R}^2$, the LSM right-hand side is

$$(\phi_z)_r = \exp(-ik\widehat{x}_r \cdot z). \tag{2}$$

Training shapes are disks and ellipses. Kites and pairs of disks are held out as geometric out-of-distribution (OOD) tests. The main configuration uses $m = 32$, $k = 8$, a $32 \times 32$ probing grid on $[-0.8, 0.8]^2$, and relative complex Gaussian noise drawn between 3% and 12% during training. Evaluation uses 8% and 15% noise.

# 3 Fixed nonlinear GP geometry

For acquisition angles $\theta_j$, the prescribed von Mises kernel and its row-normalized nonlinear attention are

$$W_{j\ell} = \exp\{\gamma \cos(\theta_j - \theta_\ell)\}, \qquad S_{j\ell} = \frac{W_{j\ell}}{\sum_p W_{jp}}. \tag{3}$$

The positive GP covariance is

$$K_g = (1-\rho)I + \rho D_W^{-1/2} W D_W^{-1/2}, \qquad \gamma = 1, \quad \rho = 0.2. \tag{4}$$

Both $\gamma$ and $\rho$ are modelling choices. They are buffers, not learned parameters or temperatures. With residual scale $\lambda_D$, Bayesian least squares gives the dual system

$$H_D q_z = \phi_z, \qquad H_D = F_D K_g F_D^* + \lambda_D I, \qquad g_z = K_g F_D^* q_z, \tag{5}$$

and the reported LSM score is

$$s_D(z) = -\frac{1}{2} \log\left(g_z^* K_g^{-1} g_z\right). \tag{6}$$

# 4 Tied in-context architecture

Jacobi scaling and a row-sum bound are applied to (5), without an eigendecomposition. Denote the scaled Hermitian positive operator by $A_D$ and its right-hand side by $b_z$. The recurrent cell is complex PCG. With $r_0 = b_z$, $u_0 = 0$ and $p_0 = P_D r_0$, it applies

$$\alpha_t = \frac{\langle r_t, P_D r_t \rangle}{\langle p_t, A_D p_t \rangle}, \qquad u_{t+1} = u_t + \alpha_t p_t, \tag{7}$$

$$r_{t+1} = r_t - \alpha_t A_D p_t, \qquad \beta_t = \frac{\langle r_{t+1}, P_D r_{t+1} \rangle}{\langle r_t, P_D r_t \rangle}, \tag{8}$$

$$p_{t+1} = P_D r_{t+1} + \beta_t p_t. \tag{9}$$

The same cell and preconditioner are reused at every depth. To incorporate the fixed nonlinear attention without corrupting Krylov conjugacy, the learned object is constrained to be SPD:

$$L_\Gamma = I - D_W^{-1/2} W D_W^{-1/2}, \qquad P_D = I + a_D L_\Gamma + b_D L_\Gamma^2, \qquad a_D, b_D \geq 0. \tag{10}$$

A small invariant controller reads six scalar summaries of $A_D$ once and returns $(a_D, b_D)$. It does not see spatial query coordinates and cannot emit an image. The exact CG coefficients in (9) are computed from the prompt $(F_D, \phi_z)$, which is the in-context computation.

The training loss is a balanced pairwise ranking loss between interior and exterior probing points plus $0.2 \log(1 + \|r_T\|/\|b\|)$. A direct solve is never used as a training target. Direct Tikhonov appears only as a held-out reference. Each of three seeds uses 1500 gradient steps, batch size 16 and AdamW at learning rate $10^{-3}$.

# 5 Fixed-geometry results

Table 1 reports 15% noise. Each entry pools 576 independent tasks (192 per seed); uncertainty is a 95% confidence interval over tasks. The eight-loop indicator is already accurate although its algebraic residual is not yet small. At 20 loops, learned PCG and the direct solve are visually and statistically indistinguishable. The learned SPD preconditioner reduces the OOD-kite mean residual from 0.830 to 0.746 at eight loops and from 0.0141 to 0.0102 at 20 loops.

**Table 1:** Average precision at 15% noise; mean ± 95% CI.

| Method | ellipse (ID) | kite (OOD) | two disks (OOD) |
|---|---|---|---|
| identity CG, 8 loops | $0.9915 \pm 0.0006$ | $0.9879 \pm 0.0008$ | $0.9135 \pm 0.0028$ |
| trained PCG, 8 loops | $0.9915 \pm 0.0007$ | $0.9881 \pm 0.0008$ | $0.9136 \pm 0.0028$ |
| trained PCG, 20 loops | $0.9917 \pm 0.0006$ | $0.9889 \pm 0.0008$ | $0.9131 \pm 0.0028$ |
| direct Tikhonov | $0.9917 \pm 0.0006$ | $0.9889 \pm 0.0008$ | $0.9131 \pm 0.0028$ |

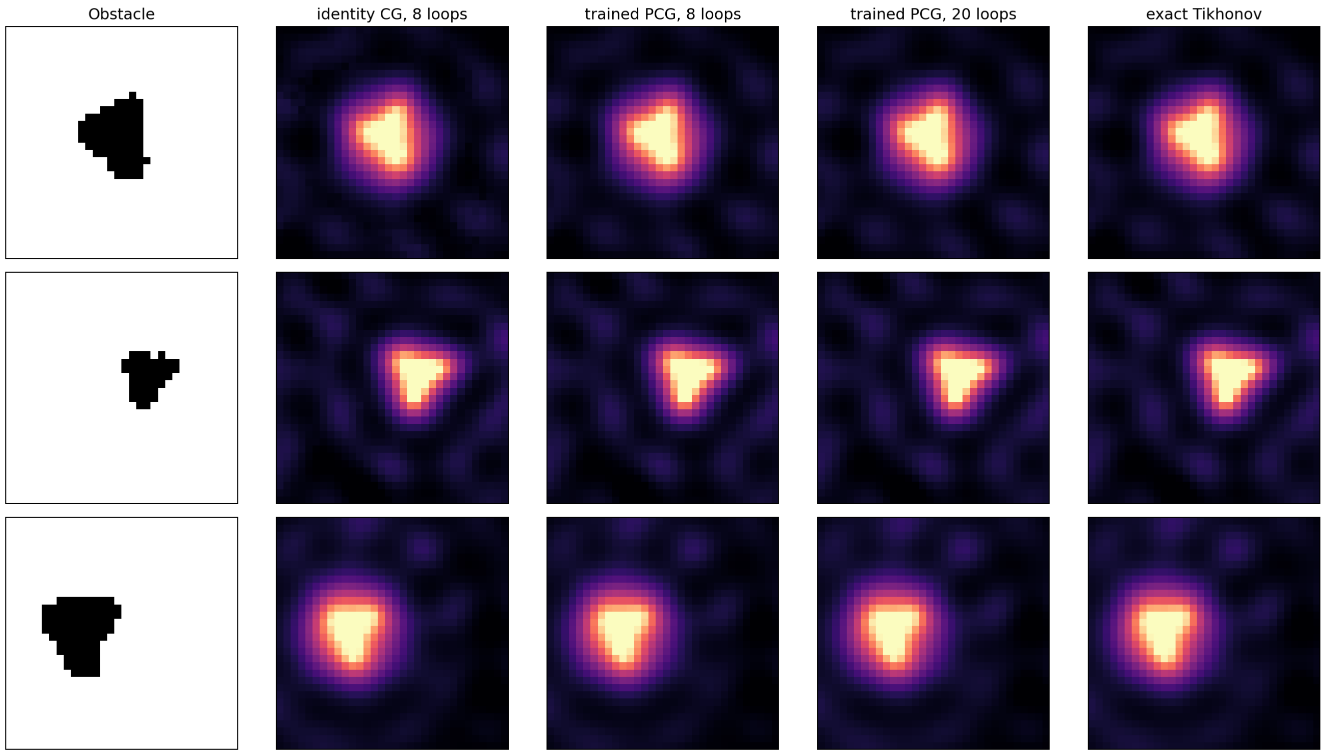


**Figure 1:** Three unseen kites at 8% noise. The output is the LSM score obtained from recurrently computed $g_z$, not a neural image prediction.

The depth audit separates image quality from solver fidelity. Average precision saturates after roughly four loops, whereas the residual continues to fall by two orders of magnitude through depth 20. On an H100, a batch of 24 matrices and all 1024 probing points takes 11.69 ms at eight loops, 15.53 ms at 20 loops, and 15.51 ms for the direct solve. Thus this small $32 \times 32$ experiment establishes computational structure, not a large-scale speed advantage.

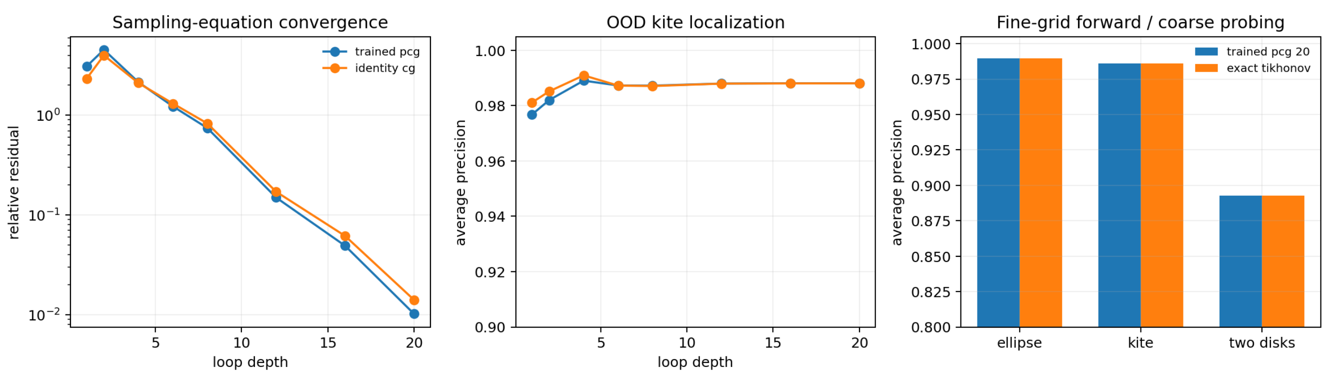


**Figure 2:** Left: convergence of the sampling equation. Middle: localization saturates before the algebraic solve. Right: the fine-grid forward control uses a $64^2$ scattering grid and a separate $32^2$ probing grid.

The separate-grid control rules out the simplest inverse crime. With data assembled on $64^2$ points but probing on $32^2$, trained PCG retains average precision 0.9896/0.9858/0.8926 on ellipses/kites/two disks, exactly matching the direct-solve ranking to four decimals.

# 6 Controlled recurrent-solver ablation

This is a small ICL experiment. The controller is a $6 \to 32 \to 32 \to 2$ MLP with 1,346 trainable parameters (5.3,kB in single precision). To isolate the effect of the iterative method, Richardson, heavy-ball, Chebyshev and PCG use exactly the same six invariant summaries of $A_D$, the same fixed GP kernel, the same Bayesian system (5), the same score decoder (6), training tasks, loss, depth, seeds and optimizer. Only the recurrent update changes. In particular, the stationary baselines take the form

$$u_{t+1}^{\mathrm{R}} = u_t + \eta_D(b - A_D u_t), \tag{11}$$
$$u_{t+1}^{\mathrm{HB}} = u_t + \eta_D(b - A_D u_t) + \beta_D(u_t - u_{t-1}), \tag{12}$$

where the controller selects stable task-conditioned coefficients. The Chebyshev semi-iteration uses the same second-order recurrence, with its coefficients generated from controller-selected spectral bounds $[\ell_D, L_D]$. Thus it learns an iteration polynomial, not a reconstruction map. PCG instead preserves exact in-context Krylov coefficients and uses the controller output only in the SPD preconditioner (10).

Table 2 exposes an important distinction. At 20 loops, heavy-ball and Chebyshev already produce good indicator rankings, but they have not solved the sampling equations: their OOD-kite residuals are 0.591 and 0.544, compared with 0.010 for PCG. Richardson fails both tests at this depth. The conclusion is therefore not that all loops are interchangeable; localization is insensitive to sizeable coefficient error, whereas Bayesian least-squares fidelity is not.

**Table 2:** Same-encoder/same-decoder comparison at 20 loops and 15% noise. Average precision (AP) and kite residual pool 576 tasks; runtime is milliseconds for a batch of 24 matrices and all 1024 probing points on an H100.

| Loop | parameters | ellipse AP | kite AP | two-disks AP | kite residual |
|---|---:|---:|---:|---:|---:|
| Richardson | 1,346 | 0.1062 | 0.1187 | 0.0470 | 0.7699 |
| heavy-ball | 1,346 | 0.9856 | 0.9838 | 0.8690 | 0.5911 |
| Chebyshev | 1,346 | 0.9802 | 0.9886 | 0.8691 | 0.5438 |
| PCG | 1,346 | 0.9926 | 0.9890 | 0.9118 | 0.0101 |
| direct Tikhonov | 0 | 0.9926 | 0.9890 | 0.9118 | $1.7 \times 10^{-5}$ |

The cost per 20-loop batch is 11.45,ms for Richardson, 10.99,ms for heavy-ball, 11.39,ms for Chebyshev and 15.01,ms for PCG. The modest PCG overhead buys much faster algebraic convergence. On held-out kites, Chebyshev reaches residual 0.214 at 48 loops and 0.064 at 80 loops; heavy-ball reaches 0.265 and 0.122. PCG reaches $9.6 \times 10^{-3}$ at 20 loops and $3 \times 10^{-4}$ at 32 loops. Richardson only reaches AP 0.956 at 80 loops, while its residual remains 0.644.

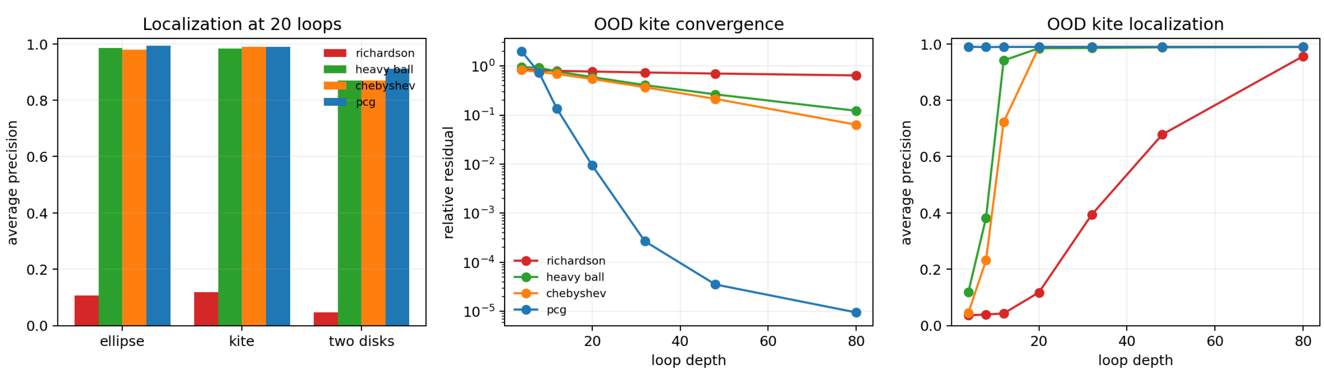


**Figure 3:** Controlled loop comparison. Left: localization at the training depth. Middle: algebraic convergence on unseen kites. Right: localization can saturate long before the sampling system is solved.

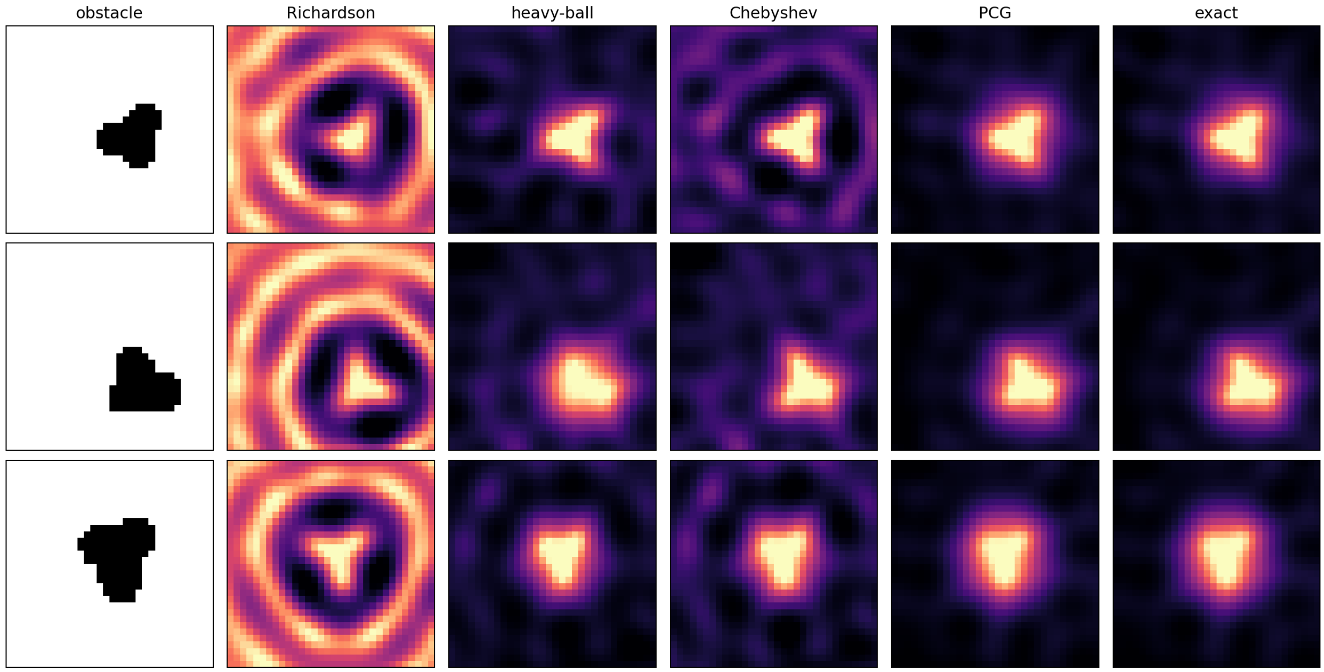


**Figure 4:** Representative unseen kites at 15% noise after 20 loops. Heavy-ball and Chebyshev localize the obstacle but retain structured artefacts; PCG is visually indistinguishable from the direct Bayesian LSM solve.

# 7 Learned experiment design

The second experiment trains six shared source/receiver angles through the frozen LSM loop. A periodic gap parameterization enforces at least 28° between distinct angles; hence repeated measurements cannot masquerade as new directions. The only optimized objective is the same LSM ranking loss. The baseline designs are six equally spaced angles and the random feasible initialization for each seed.

**Table 3:** Six-angle design at 12% noise; average precision, mean ± 95% CI.

| Design | ellipse (ID) | kite (OOD) | two disks (OOD) |
|---|---|---|---|
| learned | $0.9582 \pm 0.0037$ | $0.9628 \pm 0.0034$ | $0.8790 \pm 0.0055$ |
| random feasible | $0.8972 \pm 0.0083$ | $0.9090 \pm 0.0071$ | $0.8136 \pm 0.0088$ |
| uniform | $0.2604 \pm 0.0018$ | $0.2626 \pm 0.0018$ | $0.2439 \pm 0.0019$ |

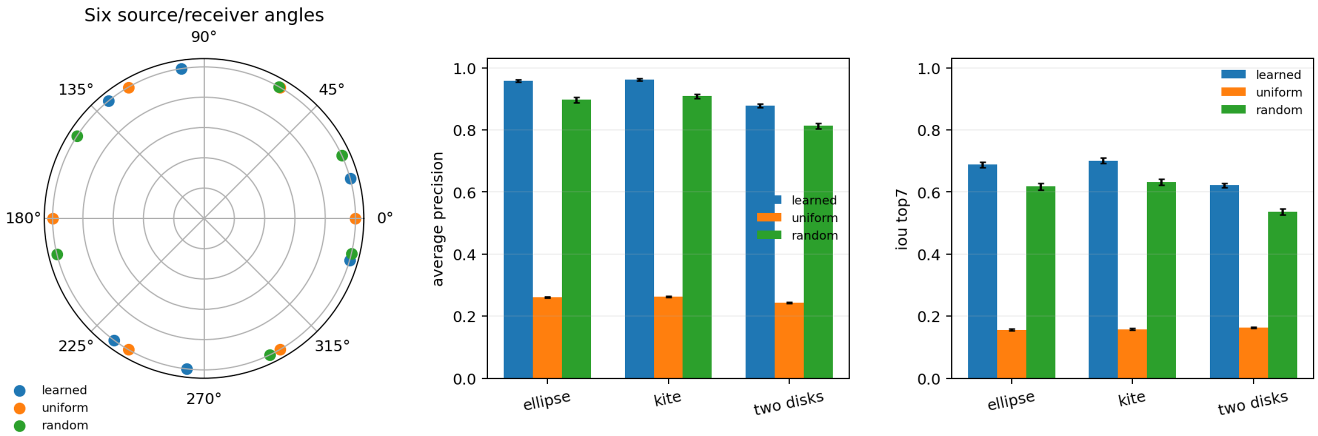


**Figure 5:** A representative learned geometry and aggregate performance over three design seeds. Independent seeds converge to rotations of a similar nonuniform paired-angle pattern.

The poor uniform baseline is not a solver failure: its six-dimensional PCG residual is below $10^{-4}$. It is an acquisition ambiguity. For the discrete multistatic wave-vector set $k(d_s - \widehat{x}_r)$, uniform spacing creates periodic replicas. The maximum point-spread sidelobe is $0.999 \pm 0.000$ for the uniform design, $0.678 \pm 0.104$ for the random design, and $0.562 \pm 0.008$ for the learned design.

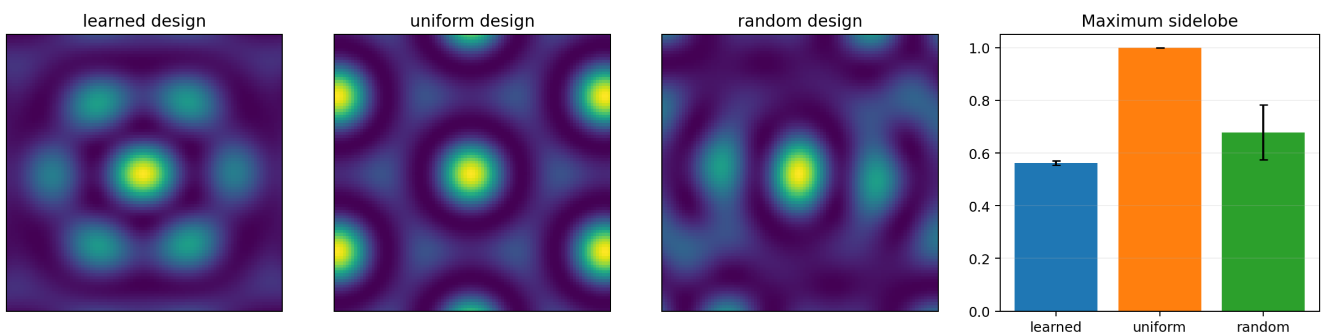


**Figure 6:** Central point-spread functions for one seed and maximum sidelobe over three seeds. The learned design suppresses the grating-lobe replicas of the uniform six-angle array.

# 8 What worked, what did not, and the bottleneck

Three conclusions are supported by the data.

1. A fixed nonlinear acquisition kernel can be embedded in a tied complex LSM loop without learning the feature geometry or an image surrogate.
2. Good-looking indicators do not imply a faithful sampling solve. Heavy-ball and Chebyshev preserve useful low-frequency localization modes at 20 loops but leave residuals two orders of magnitude above PCG. Learning arbitrary damping factors inside CG is still less safe because it also breaks Krylov conjugacy; constraining learning to the SPD preconditioner in (10) avoids that failure.
3. At this scale, the dominant gain comes from acquisition design, not from the learned preconditioner. The preconditioner improves residual convergence modestly; choosing non-aliased angles changes reconstruction quality dramatically.

The principal bottleneck is therefore no longer the reconstruction map. LSM localizes the obstacle after few Krylov steps, while faithful posterior coefficients require more depth. For a presentation, image metrics must be shown together with the sampling-equation residual; otherwise an attractive indicator can conceal an invalid solve.

# 9 Limitations and next experiment

This is a proof of mechanism, not a publishable physical benchmark. The forward map (1) is a Born–Foldy discretization, not a boundary integral solver for extended sound-soft obstacles. All data are synthetic, the wavenumber and circular aperture are fixed, and obstacle masks supervise the ranking loss. The fine-grid control removes exact grid reuse but not model mismatch. Most importantly, no claim is made yet for passive random-source LSM or for the small-random-scatterer construction. Those settings require training on $C_D$ or $\widetilde{C}_D$ while varying the number of sources, realizations, receivers and aperture. Scaling laws should be fitted only after those physically distinct experiments exist.

The next decisive test is to replace (1) by an independent sound-soft boundary-integral simulator, retain the frozen kernel and PCG cell, and evaluate the already trained loop without fine-tuning. Only then should the random-source regimes be added.

# Reproducibility

All tasks are generated online. Seeds are 17, 29 and 43. The primary run took 164 seconds and the controlled four-solver audit took 216 seconds on one NVIDIA H100 PCIe, excluding document compilation. The release contains task-level CSV files, confidence summaries, checkpoints, figures, protocol metadata, tests and the exact commands in the README.

# References


[1] J. Garnier, H. Haddar, and H. Montanelli. The linear sampling method for random sources. *arXiv:2210.15560*, 2023.

[2] J. Garnier, H. Haddar, and H. Montanelli. The linear sampling method for data generated by small random scatterers. *arXiv:2403.19482*, 2024.

[3] V. Chenu, H. Haddar, and H. Montanelli. A neural operator framework for solving inverse scattering problems. *arXiv:2602.24147*, 2026.